

Infraestructura de gas. Calidad del gas. Hidrógeno utilizado en sistemas de gas convertidos (Ratificada por la Asociación Española de Normalización en diciembre de 2023.)

UNE-CEN/TS 17977:2023

Infraestructura de gas. Calidad del gas. Hidrógeno utilizado en sistemas de gas convertidos (Ratificada por la Asociación Española de Normalización en diciembre de 2023.)

Gas infrastructure - Quality of gas - Hydrogen used in rededicated gas systems (Endorsed by Asociación Española de Normalización in December of 2023.)

Infrastructures gazières - Qualité du gaz - Hydrogène utilisé dans des réseaux de gaz convertis (Entérinée par l'Asociación Española de Normalización en décembre 2023.)

En cumplimiento del punto 11.2.5.4 de las Reglas Internas de CEN/CENELEC Parte 2, se ha otorgado el rango de documento normativo español UNE al documento normativo europeo CEN/TS 17977:2023 (Fecha de disponibilidad 2023-11-15)

Este documento está disponible en los idiomas oficiales de CEN/CENELEC/ETSI.

Este anuncio causará efecto a partir del primer día del mes siguiente al de su publicación en la revista UNE.

La correspondiente versión oficial de este documento se encuentra disponible en la Asociación Española de Normalización (Génova 6 28004 MADRID, www.une.org).

Las observaciones a este documento han de dirigirse a:

Asociación Española de Normalización

Génova, 6
28004 MADRID-España
Tel.: 915 294 900
info@une.org
www.une.org

© UNE 2023

Prohibida la reproducción sin el consentimiento de UNE.

Todos los derechos de propiedad intelectual de la presente norma son titularidad de UNE.

TECHNICAL SPECIFICATION

CEN/TS 17977

SPÉCIFICATION TECHNIQUE

TECHNISCHE SPEZIFIKATION

November 2023

ICS 75.180.01; 27.075

English Version

Gas infrastructure - Quality of gas - Hydrogen used in rededicated gas systems

Infrastructures gazières - Qualité du gaz - Hydrogène
utilisé dans des réseaux de gaz convertis

Gasinfrastruktur - Beschaffenheit von Gas -
Wasserstoff zur Nutzung in umgestellten Gassystemen

This Technical Specification (CEN/TS) was approved by CEN on 28 August 2023 for provisional application.

The period of validity of this CEN/TS is limited initially to three years. After two years the members of CEN will be requested to submit their comments, particularly on the question whether the CEN/TS can be converted into a European Standard.

CEN members are required to announce the existence of this CEN/TS in the same way as for an EN and to make the CEN/TS available promptly at national level in an appropriate form. It is permissible to keep conflicting national standards in force (in parallel to the CEN/TS) until the final decision about the possible conversion of the CEN/TS into an EN is reached.

CEN members are the national standards bodies of Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye and United Kingdom.



EUROPEAN COMMITTEE FOR STANDARDIZATION
COMITÉ EUROPÉEN DE NORMALISATION
EUROPÄISCHES KOMITEE FÜR NORMUNG

CEN-CENELEC Management Centre: Rue de la Science 23, B-1040 Brussels

© 2023 CEN All rights of exploitation in any form and by any means reserved
worldwide for CEN national Members.

Ref. No. CEN/TS 17977:2023 E

Contents	Page
European foreword	3
Introduction	4
1 Scope.....	5
2 Normative references.....	5
3 Terms and definitions	5
4 Reference conditions and pressure units	6
5 Requirements on constituents and contaminants.....	6
Annex A (informative) Reasoning for the minimum content of 98 % hydrogen	8
Annex B (informative) Development of hydrogen purity in rededicated infrastructure over time	9
Annex C (informative) Variations of Wobbe Index and Gross calorific value for hydrogen.....	10
Annex D (informative) Data from Table 1 given in alternative units.....	11
Bibliography	12

European foreword

This document (CEN/TS 17977:2023) has been prepared by Technical Committee CEN/TC 234 “Gas infrastructure”, the secretariat of which is held by DIN.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website.

According to the CEN-CENELEC Internal Regulations, the national standards organizations of the following countries are bound to announce this Technical Specification: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye and the United Kingdom.

Introduction

This document for hydrogen in rededicated gas systems takes into account the condition of existing piping with possible presence of liquid and solid deposits influencing the hydrogen quality.

Many parameters given in this document are deduced from EN 16726.

This document is the first edition and reflects the current situation, taking into account the existing grid, see Annex B. It is expected that over time the hydrogen delivered through such pipework will improve in quality. This will be taken into account in further development of the document.

This document is supposed to balance the requirements of gaseous hydrogen for the producer, the different grid operators and the end-users, respectively in order to set-up a reliable backbone of future energy supply which is (partly) based on using rededicated gas infrastructure. As a consequence, on-site pre-treatment cannot be excluded for applications and/or processes with specific stringent requirements.

The proposed parameters are verified by German project results (HyQual) which have been consulted in combination with end-users' specifications. In the project, different sources of hydrogen have been evaluated, including pyrolysis, steam reforming, electrolysis (Chlorine-Alkaline process and water electrolysis). Furthermore, the proposal takes biological hydrogen production process into account.

Also, the results of the Work Package 9 of the EC-CEN/GERG PNR project 'Removing the technical barriers to use of hydrogen in natural gas networks and for (natural) gas end users' have been taken into account.

This document supports the whole value chain (producers, system operators, other related services, end-users) to produce, transport, store, deliver and/or use hydrogen in a possibly rededicated (or adapted) natural gas infrastructure and connected applications in a safe way in the CEN member countries and where applicable, without the risk to harm or damage the infrastructure or connected user applications.

However, it is recognized that some end-users are sensitive to some gas quality parameters and impurities. In these cases, purification measures can be needed, for upgrading the hydrogen quality.

This document supports the implementation of the hydrogen strategy on European and national level and will accordingly facilitate the trade of hydrogen across entire Europe.

In the context of this document, the following aspects require close co-operation respectively co-ordination with other TCs (CEN-CLC/JTC 6, CEN/TC 268, ISO/TC 197, TCs for equipment and application):

- taking into account both natural gas systems and new built hydrogen systems;
- hydrogen qualities for fuel cell applications EN 17124:2022 (CEN/TC 268 WG 5, legally binding in some countries, e.g. DE).

1 Scope

This document defines the quality of gaseous hydrogen, i.e. its parameters and limiting values, to be transmitted, injected into and extracted from storages, distributed and utilized in fully and/or partially rededicated gas infrastructure and connected applications in a safe way.

This document gives evidence to the end-user which minimum exit hydrogen quality can be expected and ensured from natural gas infrastructure as minimum requirement and without further purification.

NOTE 1 The rededicated gas infrastructure can include new parts of this infrastructure constructed/added after the conversion of the natural gas grid.

NOTE 2 It is expected that over time the hydrogen delivered through such pipework will improve in quality, e.g. due to the increase in share of high purity hydrogen produced by electrolysis. This will be taken into account in further development of this document.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN ISO 13443, *Natural gas - Standard reference conditions (ISO 13443)*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp/>
- IEC Electropedia: available at <https://www.electropedia.org/>

3.1

rededicated gas system **repurposed gas system**

gas infrastructure originally used to convey or store gases of the first and second family which will be used from the time of rededication for other fluids as e. g. hydrogen. It can include new parts of infrastructure constructed/added after the conversion of the natural gas grid

Note 1 to entry: After a safety and reliability assessment the components of the infrastructure should be found suited for conveying hydrogen.

3.2

hydrogen

gaseous hydrogen with a certain purity as described in Clause 5

Note 1 to entry: The definition is focussing the purpose of this document.

3.3

exit point

point at which gas leaves the gas transmission or distribution system for end-use

4 Reference conditions and pressure units

Unless stated otherwise all volumes are given for the real dry gas at ISO standard reference conditions of 15 °C (288,15 K) and 1013,25 mbar (101,325 kPa). Unless otherwise stated all pressures are absolute pressures.

Whenever data on the volume, gross calorific value (GCV), energy and Wobbe Index (WI) are communicated, it shall be specified under which reference conditions these values were calculated.

In assessing compliance with this document parameters should be determined directly at ISO standard reference conditions. If the properties are only available at other reference conditions and the actual gas composition is not known, then conversion to ISO standard reference conditions shall be carried out using the procedure described in EN ISO 13443.

NOTE Besides the ISO standard reference conditions given in EN ISO 13443 (15 °C/15 °C), other reference conditions are used. Particularly in gas transmission, (25 °C/0 °C) mostly is taken as normal reference conditions.

5 Requirements on constituents and contaminants

Hydrogen transported and distributed through rededicated gas networks shall comply with the requirements given in Table 1 and shall be accepted for conveyance.

NOTE 1 The hydrogen quality requirements apply to every exit point of the gas system.

Table 1 — Quality requirements for hydrogen in rededicated gas networks

Parameter	Unit	Value	Reference standards for test methods (informative)
Hydrogen	mol-%	≥ 98 (see Annex A)	DIN 51894
Wobbe Index	MJ/m ³ (15 °C/15 °C)	42,0 – 46,0	EN ISO 6976
The content and composition of the further quality parameter (e.g. sum of inerts) shall satisfy the Wobbe Index value above.			
Water	μmol/mol	≤ 250 ≤ 60 ^a	ISO 21087
Hydrocarbon dew point (HCDP) ^b	°C	< -2 °C at 1 < p < 70 bar	ISO 21087
Sum of inerts (N ₂ , He, Ar)	mol-%	≤ 2	ISO 21087
Gaseous hydrocarbons ^b	mol-%	≤ 2	ISO 21087
Oxygen (O ₂) ^c	mol-% μmol/mol	≤ 0,1 ^d ≤ 10	ISO 21087
Carbon monoxide	μmol/mol	≤ 20	ISO 21087
Carbon dioxide	μmol/mol	≤ 20	ISO 21087
Total sulfur ^b	μmol/mol	≤ 7 ^e	ISO 21087

Parameter	Unit	Value	Reference standards for test methods (informative)
Ammonia	μmol/mol	≤ 13	ISO 21087
Halogenated compounds	μmol/mol	≤ 0,05	ISO 21087
max. particulate concentration ^b	mg/kg	technically free	ISO 21087
Contaminants	The gas shall not contain constituents other than listed in this table at levels that prevent its transportation, storage and/or utilization without quality adjustment of treatment.		ISO 21087
a 250 μmol/mol at MOP less or equal to 10 bar, 60 μmol/mol at MOP over 10 bar. b These components most likely have their source in the previous use of the pipework. c Rolling 24 h average. d Max. 0,1 mol-% in grids with no exit point to UGS or to sensitive customers, otherwise max. 10 μmol/mol. e Non-odorised hydrogen.			

NOTE 2 For convenience, Table 1 is transposed to alternative units, that the gas industry is more familiar with, in Annex D, Table D.1.

In addition to the contaminants featured in Table 1, the hydrogen shall not contain any constituents that can impede safety or the integrity of the infrastructure and/or of gas appliances and operations of end-users. Appropriate measures shall be taken.

NOTE 3 Applications are sensitive towards variation of the gas quality depending on the type of application and the degree of variation.

In particular, during the commissioning phase of the rededicated gas system it is recommended to monitor the gas quality at the exit point closely.

For sampling the general principles of EN ISO 10715 and ISO 19880-9¹ can be used for guidance.

The Wobbe Index and calorific value – depending on the variation of the constituents – can vary (see Annex C, Table C.1).

¹ Under preparation. Current stage at the time of publication: ISO/DIS 19880-9.

Annex A

(informative)

Reasoning for the minimum content of 98 % hydrogen

Hydrogen produced by electrolysis is characterized by a very high hydrogen content whereas hydrogen produced by reforming normally has a lower hydrogen content depending on the purification technique used.

The various end user segments have different requirements regarding the required minimum hydrogen concentration. For example, end users using hydrogen to decarbonise their high temperature processes or end users that operate gas engines or gas turbines can use hydrogen fuel with a lower minimum hydrogen concentration. End users using hydrogen as a chemical reactant require for some of their processes a feedstock with a high content of hydrogen.

The hydrogen backbone, connecting production and demand, can only be operated with a certain minimum hydrogen value valid for the whole backbone system. Since the scope of this document is referring to rededicated pipelines, the possible contamination of the transported hydrogen by natural gas residues present in the pipeline needs to be taken into account as well.

A review of the existing documents on the hydrogen requirements of (industrial) customers showed that very little information is available on this topic in the public domain. Detailed forecasts are missing for the expected shares of hydrogen the end-user segments will use and how the hydrogen production is subdivided into the different hydrogen production techniques.

Furthermore, most of the specifications that are currently under development or recently published to be applied to partly rededicated gas infrastructure specify a minimum hydrogen concentration of 98 mol-%. Higher purity of hydrogen with a lesser content of contaminants can be achieved at any given exit point by the means of purification units.

Annex B (informative)

Development of hydrogen purity in rededicated infrastructure over time

Over time of use of a rededicated infrastructure, the amounts of constituents and contaminants which are due to the previous use of the infrastructure will significantly decrease. However, at the time of publication of this document no time frame can be estimated. The original amounts of contaminants, their nature, the flow rate (i.e. the velocity and the total volume) of hydrogen will influence this process.

The rededication of the pipework will not be a simultaneous process everywhere, but will extend over a period of a number of years.

NOTE As an example given, sulfur from odorants will be encountered in the majority of rededicated pipe systems.

Annex C (informative)

Variations of Wobbe Index and Gross calorific value for hydrogen

Table C.1 — Variations of Wobbe Index and Gross calorific value for hydrogen according to this document, taking into account the different minor constituents under reference conditions (15 °C (288,15 K) and 1013,25 mbar (101,325 kPa))

Designation	Symbol	Unit	Reference conditions		
			15 °C / 15 °C/ 101,325 kPa		
			2 mol % methane	100 mol % hydrogen	2 mol % nitrogen
Wobbe index	$W_{s,n}$	MJ/m ³	44,81	45,88	40,09
Superior calorific value (network)	$H_{s,n}$	MJ/m ³	12,61	12,10	11,86
Relative density	d_n		0,079	0,07	0,087
EXAMPLE Values for compositions with respectively 98 mol % hydrogen and 2 mol % methane or 2 mol % nitrogen.					
NOTE Some of the following example gas qualities are off-spec gas according to Table 1.					

Annex D (informative)

Data from Table 1 given in alternative units

Table D.1 contains alternative units that the gas industry is more familiar with.

Table D.1 — Quality requirements for hydrogen in rededicated gas networks in alternative units

Parameter	Unit	Value	Reference standards for test methods (informative)
Hydrogen	mol-%	≥ 98	DIN 51894
Wobbe Index	MJ/m³ (15 °C/15 °C)	42,0 – 46,0	
The content and composition of the further quality parameter (e.g. sum of inerts) shall satisfy the Wobbe Index value above.			
Water	mg/m³	≤ 200 ≤ 50 ^a	ISO 21087
Hydrocarbon dew point (HCDP) ^b	°C	< -2 °C at 1 < p < 70 bar	ISO 21087
Sum of inerts (N ₂ , He, Ar)	mol-%	≤ 2	ISO 21087
Gaseous hydrocarbons ^b	mol-%	≤ 2	ISO 21087
Oxygen (O ₂) ^c	mol-% μmol/mol	≤ 0,1 ^d ≤ 10	ISO 21087
Carbon monoxide	μmol/mol	≤ 20	ISO 21087
Carbon dioxide	μmol/mol	≤ 20	ISO 21087
Total sulfur ^b	mg/m³	≤ 10 ^e	ISO 21087
Ammonia	mg/m³	≤ 10	ISO 21087
Halogenated compounds	mg/m³	≤ 0,08	ISO 21087
max. particulate concentration ^b	mg/kg	technically free	ISO 21087
Contaminants	The gas shall not contain constituents other than listed in this table at levels that prevent its transportation, storage and/or utilization without quality adjustment of treatment.		ISO 21087
^a 200 μmol/mol at MOP less or equal to 10 bar, 50 μmol/mol at MOP over 10 bar.			
^b These components most likely have their source in the previous use of the pipework.			
^c Rolling 24 h average.			
^d Max. 0,1 mol-% in grids with no exit point to UGS or to sensitive customers, otherwise max. 10 μmol/mol.			
^e Non-odorised hydrogen.			

Bibliography

- EN 16726, *Gas infrastructure - Quality of gas - Group H*
- EN 17124:2022, *Hydrogen fuel - Product specification and quality assurance for hydrogen refuelling points dispensing gaseous hydrogen - Proton exchange membrane (PEM) fuel cell applications for vehicles*
- EN ISO 6976, *Natural gas - Calculation of calorific values, density, relative density and Wobbe indices from composition (ISO 6976)*
- EN ISO 14532, *Natural gas - Vocabulary (ISO 14532)*
- EN ISO 10715, *Natural gas - Gas sampling (ISO 10715)*
- ISO 14687, *Hydrogen fuel quality - Product specification*
- ISO 14912, *Gas analysis - Conversion of gas mixture composition data*
- ISO 19880-9, *Gaseous hydrogen - Fuelling stations - Part 9: Sampling for fuel quality analysis*
- ISO 21087, *Gas analysis - Analytical methods for hydrogen fuel - Proton exchange membrane (PEM) fuel cell applications for road vehicles*
- DIN 51894, *Gasanalyse - Gaschromatographisches Verfahren für Brenngase/ Erdgas und andere Gasgemische Gas analysis - Gas chromatographic method for fuel gases/natural gas and other gas mixtures*
- PAS 4444:2020+A1:2021, *Hydrogen-fired gas appliances. Guide*
- Jochen Schütz, Kerstin Krüger, Dr. Tobias van Almsick, *H₂ im Netz - Voruntersuchungen zu Gasbeschaffungsaspekten bei Transport und Verteilung von Wasserstoff mit Hilfe der bestehenden Erdgasinfrastruktur (HyQual), Final project report DVGW G 201921 2021-08 Gas, DVGW-Forschungsstelle am Engler-Bunte-Institut des KIT, Karlsruhe; Open Grid Europe GmbH, Essen*
- EASEE-gas CBP 2022-001/01, *Hydrogen Quality Specification, 2021-09-14*
- Removing the technical barriers to use of hydrogen in natural gas networks and for (natural) gas end users, Final reporting, EC-CEN Specific Agreement No CEN/2019/ENER/C2/452-2019/SI2.831737, 2021
- The Hy4Heat WP2 report. <https://www.hy4heat.info/s/WP2-Report-final.pdf>